

Cosmology Research Programme:

# Investigations, Results and Pending Tests

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A five-page technical summary of the PLAMB/FR, supernova, SPARC, SU(2), Quaia, HHT, SkyMap and BBN work

Prepared for the PLAMB research programme

18 July 2026 - current evidence summary

## Research status

The programme now has reproducible tests across standard candles, galaxy rotation curves, sky anisotropy and primordial abundances. The 18 July extension also supplies a charge-conjugation identifiability boundary and independently preregistered environmental replication tests. No stream currently passes every promotion gate.

The governing claim boundaries are: SPARC subset wins, not a full-sample win; SPARC rotation curves do not intrinsically identify antimatter; the environmental branch is unconfirmed and underpowered at the earlier effect scale; Quaia is a follow-up candidate only; FR/LINX is a standard expanding-background proxy; and the SU(2)-CMB thermal benchmark is separate from the late-time SU2/SU2R fluid.

## Abstract

The research began with phenomenological tests of whether redshift-distance relations, light propagation or clock rate effects could imitate part of the standard cosmological signal.

It expanded into a SPARC optical-depth representation of the radial acceleration relation, SU2/SU2R background fits, Quaia angular scans, SkyMap and Hilbert-Huang Transform (HHT) residual searches, and finally Big Bang nucleosynthesis (BBN) calculations using LINX.

The strongest current contributions are methodological: full-covariance likelihoods, hierarchical calibration offsets, preregistered redshift frames, mock-catalogue look-elsewhere corrections, non-centred posterior sampling, charge-identifiability derivations, injection-recovery calibration, immutable result manifests and independent numerical validation.

Positive local patterns exist, but calibration, spatial clustering, predictive stress, sampler divergences, failed environmental replication and primordial-abundance incompatibilities prevent discovery language.

The next phase should emphasise independent catalogues, externally justified priors, matched likelihoods and model-specific first-principles derivations.

**Keywords:** PLAMB; FR cosmology; supernovae; SPARC; SU(2); Quaia; HHT; BBN; LINX; reproducibility.

# 1. Supernovae, Calibration and Redshift Frames

**Scope.** The standard-candle programme uses Pantheon+SH0ES, DES-SN5YR/Dovekie and Union3.1 distance moduli with their published covariance or precision matrices.

The principal code is `run_joint_rawmu_hierarchical_offset_likelihood.py`, supported by `plamb_rawmu_model_utils.py` and the July 16 no-loss FR/LCDM follow-up programmes.

The promoted diagnostic is the hierarchical raw-MU likelihood, not a free per-catalogue shift.

## Core hierarchical and multiplicative-frame definitions

$$\begin{aligned}
 5 \log_{10}[(c/H_0) z_i (1 + \beta z_i)] + 25 + \delta_d(i) + \mu_i &= \delta_s(i) \\
 \delta_d &= \text{Normal}(0, \sigma_{\text{dataset}}^2) \\
 \delta_s &= \text{Normal}(0, \sigma_{\text{survey}}^2) \\
 1 + z_{\alpha} &= (1 + z_{\text{HEL}})[(1 + z_{\text{HD}})/(1 + z_{\text{HEL}})]^{\alpha}
 \end{aligned}$$

**Table 1. Supernova tests completed through 16 July 2026**

| Completed test       | Principal readout                                                                              | Standing interpretation                                                      |
|----------------------|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Joint hierarchy      | N = 3,422; moderate dataset+survey row beta<br>= 0.521314 at $H_0 = 67.5$                      | Beta remains close to 0.5, but the improvement is calibration-mediated.      |
| Offset prior audit   | Pantheon+ HD dataset pull about -2.71 sigma under moderate priors; >5 sigma under tight priors | Replace generic grids with release-based calibration budgets.                |
| Preregistered frames | Pantheon+: zHD primary; DES: zHD primary; zHEL/zCMB controls                                   | zHD is the default external-flow readout.                                    |
| Frame crossing       | Exact-log AIC preference disappears by $\alpha = 0.4739$ (Pantheon+) and 0.7779 (DES)          | The earlier heliocentric preference is redshift-frame sensitive.             |
| FR no-loss ladder    | HD FR beats free- $\Omega_m$ LCDM by base BIC in 4/4 budgets, posterior objective in 3/4       | Favourable but not robust to calibration pulls or catalogue/redshift splits. |
| Follow-up gates      | Union3.1-only and $z \geq 0.5$ : 0/4 BIC wins and 0/4 posterior wins                           | High-redshift and independent-catalogue support is presently absent.         |

## 1.1 Current conclusion

The raw-MU branch remains a useful calibration diagnostic and an economical low-to-mid-redshift phenomenology. It is not yet a cosmological detection.

Promotion requires externally documented dataset and survey calibration budgets, stable preference in leave-one-dataset-out and high-redshift tests, and a same nuisance comparison with Lambda-CDM.

zHD remains the headline frame; zHEL and zCMB remain controls unless a distinct physical model preregisters another choice.

## 2. Galaxy Acceleration and Angular Structure

### 2.1 SPARC optical-depth branch

Optical-depth representation of the radial acceleration relation

$$\begin{aligned}\tau &= [g_{\text{bar}}/(\kappa c H_0/(2\pi))]^p \\ \nu &= [1 - \exp(-\tau)]^{(-1)} \\ g_{\text{obs}} &= \nu g_{\text{bar}}\end{aligned}$$

The filtered all-Q2 family contains 157 galaxies and 3,001 rotation points after removing six preregistered persistent-negative systems. The four-chain non-centred NUTS run achieved maximum rank  $\hat{R}$  1.00259 for RAR and 1.00283 for PLAMB, but failed the locked zero-divergence criterion with 3 and 6 divergences respectively.

The fitted PLAMB exponent remains near one half ( $p = 0.51892 \pm 0.00494$ ), but the all-Q2, low-acceleration outer-point and inclination predictive gates remain failures. The standing conclusion is subset wins, not a full-sample win.

Complete FR charge conjugation swaps the matter and antimatter background contributions while preserving total background magnitude and  $|\chi|$ . The implemented rotation-curve likelihood is therefore charge-even and cannot classify an isolated SPARC galaxy as matter or antimatter.

#### SPARC/FR gate

The preregistered 2MRS/NED environment test found a suggestive negative partial association in 66 galaxies ( $\rho = -0.3073$ ,  $p = 0.01180$ ), but predictive improvement was only 2.40%; the locked development gate failed and all five replication galaxies stayed sealed.

An independently published Kourkchi-Tully group-catalogue predictor was null in 65 galaxies ( $\rho = 0.04768$ ,  $p = 0.7044$ ) and worsened held-out RMSE by 5.41%. Injection-recovery gives only 36.50% empirically recalibrated joint power at  $|\rho| = 0.307254$ ; approximately  $|\rho| = 0.5$  is needed for at least 80% recovery.

### 2.2 Quaia/SU2 angular tests

Controlled angular residual model: dipole plus survey templates

$$r_i = a + d \cdot n_i + \sum_k \gamma_k T_{ik} + \epsilon_i$$

A redshift-window candidate near  $z = 1.0$ -1.5 survived several object-level checks, but the full all-window look elsewhere probability was ordinary ( $p_{\text{global}}$  about 0.43 for the original strongest row).

After catalogue-quality matching the targeted family had  $p = 1$ .

Spatial-block mocks gave family  $p = 0.0135$ , 0.0923 and 0.2326 for 8, 12 and 16 degree cells, so the mode fails the requirement to remain below 1% across cluster scales.

The locked direction has template  $R^2 = 0.1636$  and variance-inflation factor 1.196, with strongest correlation -0.4044 to a Gaia scan-angle template.

Injection-recovery exposed an important refinement: at the observed 1x amplitude, strict physical recovery after scan-law and colour control was 0.2833, while the final BIC gate accepted only 0.0011; 0.2823 of trials were physically recovered and rejected only by BIC.

The current template gate is therefore under-powered and collinear as well as non-confirmatory. The correct status is targeted follow-up candidate only, with no SU2 detection or headline anisotropy claim.

### 2.3 SkyMap and HHT archive

The historical SkyMap/Quaia and CMB-supernova HHT work has been curated with code, extracted reports and representative figures.

HHT identified intrinsic-mode-function structure, but shuffled-order controls were ordinary: extended SU2-minus-Lambda-CDM maximum and summed-energy upper-tail probabilities were 0.9703 and 0.9604.

Archive this as method development and candidate-feature evidence; a new preregistered probe separated analysis is required.

### 3. Primordial-Abundance Tests

#### 3.1 FR/LINX validation

The FR investigation was translated into a standard expanding-background LINX clock/radiation proxy and scanned over 30,503 immutable key-network states.

A preregistered 92-evaluation validation matrix then tested  $\text{rtol} = 10^{-7}$ ,  $\text{atol} = 10^{-11}$ , 300/600 interpolation samples, key/small/full PRIMAT 2023 networks, updated neutron-lifetime states, the three main deuterium-sensitive rates and five omitted Li/Be channels.

All 92 evaluations completed with no missing or failed rows.

##### Numerical and local deuterium-theory readout

$$\begin{aligned}\max |\Delta(10^5 \text{ D/H})| &= 5.51 \times 10^{-4} \\ \max |\Delta Y_p| &= 6.53 \times 10^{-5} \\ \max |\Delta(10^{10} \text{ Li/H})| &= 1.71 \times 10^{-3} \\ \sigma_{\text{D,theory}} &= 0.02608\end{aligned}$$

at the strict full-network baseline

**Table 2. Publication-grade FR/LINX validation rows**

| Strict full-network state | $10^5 \text{ D/H}$ | $Y_p$     | $10^{10} \text{ Li/H}$ | Registered $\chi^2$     |
|---------------------------|--------------------|-----------|------------------------|-------------------------|
| Baseline                  | 2.434914           | 0.2469325 | 5.637694               | 295.6 (reference scale) |
| Best registered direct-Li | 2.585689           | 0.248639  | 4.647982               | 210.57                  |

There are zero registered and zero updated-current simultaneous D+He+Li two-standard-deviation passes.

Full minus-key corrections are at most 0.000649 in D,  $1.48 \times 10^{-5}$  in helium and 0.0580 in lithium.

The largest omitted channel lithium response, Be7daap, is 0.06079 at baseline, far below the residual excess.

The main statistical correction is instead the D theory budget: 0.02608 is about 90% of the registered observational error and must be marginalised jointly with rate pulls.

#### 3.2 Literature-defined SU(2)-CMB bridge

##### High-temperature SU(2)-CMB benchmark

$$\begin{aligned}n_{\text{u,CMB}} &= 4^{(-1/3)} &= 0.629960525 \\ H_{\text{SU2CMB}}/H_{\text{SBBN}} &= 1.25326 \text{ at } T_{\text{gamma}} = 1 \text{ MeV} \\ N_{\text{eff,SU2CMB}} &= 20.373 \text{ (LINX diagnostic)}\end{aligned}$$

The separately registered 62-point bridge propagates the published high-temperature SU(2)-CMB equation of state through LINX while retaining standard weak and nuclear microphysics.

At fixed reference baryon density it predicts  $\text{D/H} = 0.64441 \times 10^{-5}$ ,  $Y_p = 0.3165$  and  $\text{Li/H} = 26.476 \times 10^{-10}$ .

The best D+He row at eta factor 0.48 still has  $\chi^2_{\text{D+He}} = 501.41$ ; the best row inside the published baryon prior has total  $\chi^2 = 3602.7$ .

No abundance pass occurs.

This is not the repository's late-time SU2/SU2R fluid and supplies no first-principles gauge-matter changes to masses, binding energies, Q-values or rates.

## 4. Pending Tests and Research Path

The next programme should be narrower, preregistered and designed around independent falsification.

The following order gives the greatest gain in research standing per unit computation.

### 1. Quaia cross-catalogue validation

Repeat the  $z = 1.0-1.5$  search in an independent quasar catalogue with mask, windows, latitude cuts and spatial block null fixed in advance.

Promotion requires a preregistered global  $p$  across sensible block scales, stable amplitude/direction, and a coherent  $l = 1-3$  pattern.

### 2. Injection-calibrated angular likelihood

Replace the fixed  $\Delta\text{-BIC} < -10$  gate with a power-calibrated rule. Jointly model dust, depth, Gaia scanning geometry, WISE colour and catalogue-quality templates.

The SU2 scalar-residual relation must beat template-only prediction on held-out sky regions.

### 3. Supernova calibration-budget hierarchy

Derive dataset and survey priors from release covariance and documented zero-point budgets. Keep zHD primary.

Repeat same-nuisance FR versus Lambda-CDM comparisons with survey, dataset and high- $z$  hold-outs; a robust result must stay within the external budget.

### 4. SPARC prediction, identifiability and power gate

Retune both non-centred models to zero divergences, then run grouped galaxy-held-out prediction. Keep matter/antimatter labels prohibited unless a signed external field and an FR-native response law are derived before data access.

Do not unseal the five reserved galaxies. First run sample-size scaling and a genuinely independent rotation-curve/environment design capable of at least 80% joint power near  $|\rho| = 0.307254$ , with an empirically calibrated null.

### 5. BBN hierarchy and first-principles models

Fit baryon density, neutron lifetime, key rate pulls and stellar lithium depletion with matched CMB/abundance likelihoods, then replicate independently.

For FR, derive non-expanding  $T(t)$ ,  $\rho(T)$ ,  $c(T)$ , masses,  $Q_{np}$ , binding energies and rates.

Keep SU(2)-CMB separate without an action-level bridge.

### 6. HHT/SkyMap confirmatory design

Define probe-separated residuals, frequency bands and sky masks before looking at the data; use phaser randomised and order-preserving surrogates, global correction and held-out redshift ranges.

The existing archive is a hypothesis generator, not confirmatory evidence.

## 4.1 Reproducibility and evidence

Code, small results, reports, environment locks and manifests are curated at [github.com/cliveboyd/Cosmology](https://github.com/cliveboyd/Cosmology) under `github_export/`.

Large inputs and chains remain local but are checksum-referenced.

Future papers should cite commit hashes, preregistrations, programme versions, dataset releases and claim gates.

Evidence index: [1] raw-MU hierarchy/frame audit (14 Jul); [2] SPARC full NUTS (16 Jul); [3] Quaia spatial-block audit (16 Jul); [4] FR/LINX matrix; [5] SU(2)-CMB bridge (17 Jul); [6] FR charge, environmental replication and power calibration (18 Jul). Full paths: `github_export/results/README.md`.

### Conclusion

The programme now has a defensible map of survival and failure.

Highest-value work is independent Quaia validation, an externally budgeted supernova hierarchy, zero-divergence held-out SPARC prediction, a powered independent SPARC environment test, and matched or first-principles BBN tests.